

Forecasting Daily Sales of Coffee Shop Products: A Comparative Study of Moving Average, ARIMA, and SARIMA

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Abstract—The coffee shop industry in Indonesia has experienced rapid growth, with local brands like Kopi Janji Jiwa and Kopi Kenangan expanding aggressively. However, coffee shops often face challenges in demand forecasting, leading to inventory mismanagement and operational inefficiencies. This study focuses on forecasting daily sales of Ice Afecto, a popular coffee shop product, using time series models to enhance business decision-making. This study compares the performance of three forecasting methods Moving Average (MA), Autoregressive Integrated Moving Average (ARIMA), and Seasonal ARIMA (SARIMA) to identify the most accurate model for short-term sales prediction. Utilizing transaction data from a coffee shop, processed and analyzed to capture daily sales trends. Results indicate that ARIMA (0,0,1) (0,0,1) outperforms both MA and SARIMA, achieving the lowest Mean Absolute Error and Root Mean Squared Error. While MA provides simplicity, it fails to capture rapid fluctuations, whereas SARIMA underperforms due to weak seasonal patterns in the data. The study highlights ARIMA's effectiveness in optimizing inventory management, workforce scheduling, and promotional strategies, offering practical benefits for small-to-medium coffee shops. This research contributes to the industry by demonstrating how data-driven forecasting can minimize waste, improve resource allocation, and enhance profitability. Future work should incorporate external factors such as weather and promotions to further refine predictions. The findings provide valuable insights for coffee shop operators seeking to implement predictive analytics in their daily operations.

Keywords—coffee shop industry, time series analysis, ARIMA, inventory optimization, business intelligence

I. INTRODUCTION

Coffee is a vital plantation commodity for Indonesia, as this commodity serves as a small-scale farmers' income source, offers industrial raw materials, plays a role in rural development for sustainable agriculture, and contributes considerably to export earnings [1]. Right now there are now plenty of prospects for business owners to open coffee shops thanks to the expansion of the tourism industry and the rising trend of coffee consumption [2]. The coffee shop industry has become an integral part of modern urban life. Today, coffee shops serve as hubs for social interaction, entrepreneurship, and community development. With the rise of coffee enthusiasts ranging from teenagers to the elderly, along with the growing demand for specialty coffee and the desire for unique and personalized experiences, the coffee shop industry has evolved to meet consumers' ever-changing preferences[3]. Currently, data-driven business decision-making in modern operations has propelled data analytics and business intelligence, along with performance management solutions,

to become fundamental organizational tools for strategic marketing success [4].

The figure demonstrates the rapid growth of Indonesia's coffee shop industry, with Kopi Janji Jiwa and Kopi Kenangan having the highest number of outlets in 2022 - even surpassing global chains like KFC and Pizza Hut. This reflects both the rising trend of coffee consumption and the aggressive expansion of local brands.

Coffee shops frequently face challenges in predicting product demand, which can lead to inventory management issues such as overstocking or stockouts. This problem becomes increasingly complex due to daily sales fluctuations influenced by various variables, including weekday/weekend patterns, seasonal factors, and the impact of promotions or discounts. These conditions create a need to identify more sophisticated and accurate forecasting models that can assist coffee shop operators in both day-to-day operational decision-making and developing more effective long-term business strategies. Proper implementation of forecasting systems will enable management to optimize raw material inventory, schedule staff shifts more efficiently, and plan promotions more precisely based on actual sales data.

This study focuses on forecasting daily sales of Ice Afecto, a popular product, using time series models. The objectives are:

- To analyze daily sales patterns of Ice Afecto.
- To compare the performance of Moving Average, ARIMA, and SARIMA models.
- To determine the best forecasting model for supporting business decisions.

The choice between Moving Average (MA), ARIMA, and SARIMA depends on data characteristics. MA works best for simple trend smoothing, ARIMA handles non-seasonal trends, while SARIMA captures seasonal patterns. Comparing these models is crucial because businesses need to balance accuracy with practicality - complex models may overfit while simple ones may underfit. For inventory management and staffing decisions, selecting the right model ensures cost-effective and reliable forecasts. This study helps small businesses choose the most suitable forecasting method for their specific operational needs. The research contributes to both academia and industry by providing a detailed analysis of sales patterns and evaluating forecasting methods tailored for small to medium-sized coffee shops.

II. LITERATURE REVIEW

A. Time Series Forecasting in Business

Sales forecasting is crucial in the food and beverage industry, enabling businesses to optimize inventory and refine marketing strategies. Time series forecasting, a key research area in data science, involves preprocessing, sequence modeling, prediction, and aggregation [5]. Large observational time-series data sets are usually accessible in these domains, but large-scale studies to ascertain causal effects are generally difficult or unattainable. Therefore, creating statistical techniques that use large datasets to determine causal links is a crucial issue [6]. This method analyzes historical patterns to predict future trends while accounting for temporal factors. Statistical approaches (e.g., ARIMA) perform well with limited data, whereas machine learning (e.g., LSTM) excels with complex patterns in large datasets.

B. Arima Model in Forecasting

Recent studies highlight the effectiveness of ARIMA models in forecasting perishable goods sales, particularly in the agricultural and retail sectors. For instance, research on vegetable sales demonstrates that seasonal ARIMA, combined with first-order differencing, can accurately predict weekly replenishment needs by accounting for trends and seasonal fluctuations [7]. This approach is especially valuable for perishable items like vegetables, where demand variability and loss rates significantly impact pricing and inventory decisions.

C. Optimizing Inventory Management with ARIMA Forecasting

Traditional time series models, such as ARIMA, have been widely employed for midterm daily load forecasting due to their ability to capture temporal dependencies and trends in historical data. Recent advancements include hybrid approaches like Wavelet-ARIMA, which decomposes load data into frequency components to enhance forecasting accuracy. Studies demonstrate that wavelet decomposition improves model performance by isolating underlying patterns and reducing noise [8]. However, these models often overlook external factors, such as climatic conditions, which can significantly influence load demand.

This methodological limitation similarly applies to business forecasting contexts like coffee shop sales, where conventional time series models (e.g., Moving Average, ARIMA) frequently fail to account for critical external variables such as weather patterns, local events, or promotional activities that directly impact daily customer footfall and product demand. Just as power systems require integrated approaches combining climatic data with load patterns, coffee shop sales forecasting would benefit from models that incorporate both internal sales trends and exogenous business environment factors

III. METHODOLOGY

A. Data Collecting and Processing

The forecasting model's accuracy has a significant influence on the end product's performance reliability and design optimization efficiency in addition to directly affecting simulation precision[9]. The dataset used in this research was obtained from a coffee shop's transaction data, consisting of two sources: training data covering historical transactions and

test data covering sales in January 2025. The data was imported in Excel format and processed using pandas. The *Date* column was converted to datetime format to ensure time-based analysis could be performed correctly. Subsequently, the data was filtered to only include transactions for the product *Ice Afecto*, which is one of the coffee shop's flagship products. Sales data confirms Ice Afecto as a flagship product, demonstrating the highest transaction volume among all menu items (Figure 3).

After filtering, the data was grouped by date to obtain the total daily sales. The data was then sorted chronologically, and the time index was set with a daily frequency to ensure data continuity. If there were dates without transactions, the sales values were filled with zero to avoid inconsistencies in the time series analysis.

B. Timeseries Forecasting Models

To forecast daily sales, three main methods were used :

- Moving Average (MA)

The Moving Average method was employed as a baseline forecasting approach. This model calculates the average sales over the last 7 days to predict future sales. While simple and easy to implement, its limitation is its inability to effectively capture seasonal patterns and long-term trends.

- ARIMA (Auto Regressive Integrated Moving Average)

The ARIMA model was used to capture trend patterns and relationships within the time series data. The optimal ARIMA parameters were determined using auto rima, which automatically selects the best values by minimizing prediction errors. This model was trained on the historical data and then used to forecast sales during the test period.

- SARIMA (Seasonal ARIMA)

The SARIMA model is an extension of ARIMA that incorporates seasonal patterns in the data. Since coffee shop sales may exhibit weekly variations, the SARIMA model uses a seasonal component with a 7-day period ($m=7$) to capture recurring patterns in daily sales. Similar to ARIMA, the optimal SARIMA parameters were selected using auto Arima before the model was trained and applied for forecasting.

C. Model Evaluation and Comparison

To assess prediction accuracy, the three models were compared using Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE). The evaluation was conducted by calculating the prediction errors against actual sales data from the first 7 days of January 2025.

D. Visualization of Forecast Results

The forecasted results were compared with actual sales through graphical representations to facilitate visual analysis. The graphs illustrate how each model follows sales trends and the extent of their deviations from actual values. These visualizations were used to determine the most suitable model for implementation in the coffee shop business.

E. Implementation for Business Decision-Making

Based on the evaluation and visualization results, the best-performing model can be adopted by the coffee shop to support inventory management and marketing strategies. Accurate forecasting enables management to prevent stock shortages or surpluses and optimize staffing levels based on projected daily demand.

IV. RESULT AND DISCUSSION

Using a variety of data aggregations and visualizations, the transaction data analysis was carried out to find sales trends. Date columns were converted, data was grouped by month, and basic information, such as statistical summaries and missing values, was displayed once the dataset was imported and processed. Investors have historically found it difficult to predict stock values because of volatility, noisy data, and sudden price swings during unstable economic times. This topic is relevant since it is necessary for both individual investors and, more significantly, for institutions and organization[10].

Histograms and box plots were used to look at sales distributions and find outliers, while line charts were used to evaluate monthly sales trends and find seasonal patterns. To determine how each product category and payment method contributed to overall sales, more research was done on them. Best-selling items were also determined in order to create more successful marketing plans.



Fig. 1. Graphic of fluctuations in one year.

This graphic displays the fluctuations in total sales by month for the year 2024. It is evident that there are significant variations in the penjualan pola, with the highest point being in Mei 2024. Penurunan penjualan yang cukup tajam is evident in the months of March and May, and is then followed by fluctuating pola until the end of the year. Mei's dramatic increase may indicate the presence of a certain factor, promotion, or, conversely, an increase in demand. Further analysis is necessary to understand the primary cause of this fluctuation.

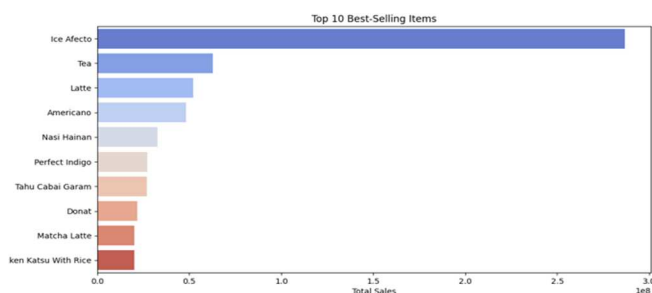


Fig. 2. Graphic of best selling products in coffeshop.

These graphic displays ten products with the highest overall sales. The first feature of Ice Afecto is its extremely low prices when compared to other products, which indicates that this product has very high demand. Other products, such as tea, lattes, and Americanos, also have significant sales figures that show consumers' preferences as compared to food. Product reviews for foods like Nasi Hainan and Chicken Katsu with Rice in the table indicate that even if they are not very substantial, some food segments do have a role in the process.

The results of the model evaluation are shown in the following table:

TABLE I. TABLE OF RESULT

Model	MAE (Mean Absolute Error)	RMSE (Root Mean Squared Error)
<i>Moving Average (7 day)</i>	264,775.51	322,972.04
<i>ARIMA (0,0,1) (0,0,1) [7]</i>	235,034.67	302,187.45
<i>SARIMA (0,0,1) (0,0,1,7)</i>	516,775.20	572,873.02

The results of the study indicate that the ARIMA (0,0,1) (0,0,1)[7] model with an intercept is the best model based on parameter optimization. With the help of the seasonal component (period in 7 day), this model stabilizes the autoregressive pattern. According to an evaluation conducted on January 7, 2025, the ARIMA model has better performance when compared to the Moving Average and SARIMA models, with an MAE of 235,034.67 and an RMSE of 302,187.45. With a larger error (MAE: 264,775.51, RMSE: 322,972.04), the Model Moving Average indicates that the 7-day rata-rata approach is not very effective at reducing data variability. Conversely, the SARIMA model, which is obviously better in estimating seasonal, just indicates the worst performance (MAE: 516,775.20, RMSE: 572,873.02). This suggests that the musical elements included in the SARIMA model may not be entirely appropriate or that the study period does not have strong musical characteristics. Accordingly, the ARIMA model is more effective than other models in estimating the *Ice Afecto* in forecasting models.

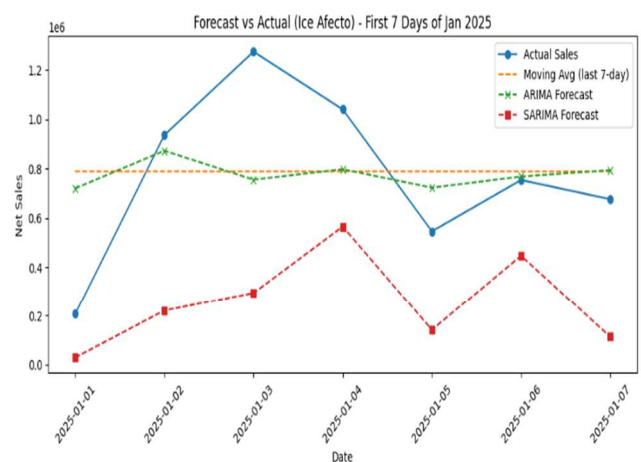


Fig. 3. Forecasting result.

A. Forecast Results Visualization

The graph compares actual sales of Ice Afecto with forecasted results from three methods: 7-day Moving Average

(MA), ARIMA, and SARIMA for the first 7 days of January 2025.

1) Actual Sales Trend

The Actual Sales (blue line) showed a significant increase from *January 1–3*, peaking on *January 3* before declining until *January 5* and slightly recovering on *January 6*. This pattern indicates sharp daily fluctuations, likely influenced by external factors such as promotions or sudden changes in customer demand.

2) Moving Average Performance

The MA (orange dashed line) produced a relatively flat forecast, failing to capture rapid trend changes since it only relies on the previous 7-day average. As a result, its predictions were less accurate compared to ARIMA.

3) ARIMA Performance

The ARIMA model (green line with cross markers) performed better than MA, with predictions closer to actual trends, particularly on *January 2 and 6*. While it still missed the sharp spike on *January 3*, ARIMA overall demonstrated superior performance in tracking fluctuations.

4) SARIMA Performance

The SARIMA model has been widely employed recently because to its increased versatility[11]. The SARIMA model (red line with square markers) significantly underestimated actual sales throughout the period. Its failure to align with historical seasonal patterns aligns with earlier evaluation results, where SARIMA recorded the highest MAE and RMSE among all models.

B. Business Implications for the Coffee Shop

Currently, almost all information is conveyed and stored in information system technology, such as menu information, discount information, and transaction data [12]. In *Afecto Coffee*, we implement it in:

1) Optimal Forecasting Model Selection

For daily operations, the coffee shop's management can adopt ARIMA to forecast demand for high-volume products like *Ice Afecto*. For example: The purchasing team can run ARIMA forecasts weekly to estimate raw material needs (milk, coffee beans, syrup), minimizing overstocking or shortages

2) Limitations of Seasonal Models (SARIMA)

Since SARIMA underperformed in predicting weekly patterns, the operations team must account for external factors:

- Daily promotions (e.g., discounts, events)
- Peak demand periods (e.g., weekends, collaborations, or community events)
- Real-time adjustments to inventory based on observed trends

3) Inventory & Stock Management

An essential component of overall corporate policy is inventory management. Its primary objective is to reduce the expenses associated with the delivery, storage, and transportation of the resources utilized in the company's product manufacturing[13]. By leveraging ARIMA's accurate demand predictions, the coffee shop can significantly enhance its inventory management practices. The forecasting model

enables precise calculation of daily ingredient requirements, particularly for perishable items like milk that need frequent replenishment to maintain product freshness. For high-variability products such as *Ice Afecto*, management can establish intelligent safety stock levels that balance the risk of stockouts against potential waste. This data-driven approach to procurement ensures optimal inventory turnover while minimizing spoilage and unnecessary costs. The system allows for dynamic adjustment of purchase orders based on predicted sales patterns, creating a responsive supply chain that adapts to anticipated demand fluctuations. This results in reduced food waste, improved cost efficiency, and consistent product availability - all critical factors in maintaining both profitability and customer satisfaction in the competitive coffee shop market.

4) Workforce Scheduling

By leveraging sales trend analysis, coffee shop managers can optimize staff scheduling to align with fluctuating customer demand. The data reveals predictable patterns, such as increased weekend traffic, allowing for proactive scheduling of additional baristas during these peak periods. To accommodate unpredictable surges, a flexible staffing model incorporating part-time workers can be quickly activated. This strategic approach to workforce planning ensures adequate coverage during busy times while maintaining cost efficiency during slower periods by avoiding overstaffing. The result is a balanced operation that delivers consistent service quality while controlling labor expenses, ultimately enhancing both employee productivity and customer satisfaction. This data-driven methodology transforms raw sales numbers into actionable staffing insights that benefit the entire business operation.

5) Promotion & Sales Strategies

By leveraging sales forecast data, the coffee shop can develop precisely timed promotional campaigns to optimize revenue throughout different periods. The predictive insights enable strategic implementation of morning combo discounts (e.g., 8-10 AM specials) to stimulate off-peak sales, while seasonal campaigns can be deployed in anticipation of forecasted demand slumps. For periods showing predicted sales spikes, limited-time specialty offers can be introduced to maximize revenue potential. This targeted approach to promotions ensures marketing resources are allocated efficiently, converting predictive analytics into tangible business results by addressing both anticipated slow periods and peak opportunities. The methodology creates a balanced promotional calendar that aligns with customer demand patterns while maintaining profit margins.

6) External Factor Analysis

While ARIMA provides reliable baseline forecasts, truly accurate demand prediction requires accounting for external variables like weather patterns, social media trends, and local events. The coffee shop should implement real-time monitoring systems to track these factors and gradually incorporate them as predictive variables in future model upgrades. This comprehensive approach bridges the gap between statistical forecasting and real-world business conditions, enabling more responsive inventory and staffing decisions when unexpected demand fluctuations occur.

7) Continuous Improvement

To ensure ongoing forecast accuracy, the coffee shop should implement a structured improvement cycle. This involves conducting monthly performance reviews that compare predicted versus actual sales figures, identifying any significant variances. When deviations exceed established thresholds, the model should be refined through advanced analytical techniques or by incorporating new predictive variables. Parallel to technical upgrades, comprehensive staff training programs should be implemented to cultivate data literacy across all operational levels, embedding a data-driven decision-making culture throughout the organization. This holistic approach maintains model relevance amidst changing market conditions while maximizing the business value derived from forecasting insights.

V. CONCLUSION

This study confirms ARIMA as the optimal forecasting model for Ice Afecto sales, delivering superior accuracy (MAE: 235,034.67; RMSE: 302,187.45) compared to Moving Average and SARIMA approaches. The model effectively captures daily sales patterns, enabling coffee shops to optimize inventory orders for perishable ingredients and implement dynamic staffing schedules. These operational improvements directly address key challenges in waste reduction and labor cost management.

The research provides an actionable framework for data-driven decision making in Indonesia's growing coffee shop industry. While ARIMA serves as a reliable foundation, managers should complement forecasts with real-time market observations and periodic model updates. The methodology balances technical feasibility with business impact, making it particularly valuable for small-to-medium coffee shops seeking to enhance operational efficiency through predictive analytics. Future enhancements could integrate weather and promotional data to further refine predictions.

This study has several limitations that should be acknowledged. First, the analysis relies solely on transaction data from a single coffee shop, which may limit the generalizability of findings to other establishments with different operational contexts. Second, missing sales dates were filled with zeros without investigating potential underlying causes, which could introduce bias. Third, the models don't incorporate external factors like weather conditions or local events that likely influence daily sales.

A. Open Data Statement:

The dataset used in this research contains sensitive financial and operational information from a business partner and is subject to confidentiality agreements. Therefore, the raw data cannot be shared publicly.

B. Author Contribution

Aulia Azzahra: Conceptualization, Methodology, Formal Analysis, Investigation, Data Curation, Writing – Original Draft, Visualization.

Ivan Diryana Sudirman: Supervision, Software, Validation.

REFERENCES

- [1] A. Tresliyana Suryana, Y. Saleh, T. Gita Dewi, and H. Sulistyawati Purwaning Rahayu, "Global competitiveness of coffee products: A comparative study of Indonesia and Vietnam," *Coffee Sci*, vol. 19, pp. 1–11, 2024, doi: 10.25186/v19i.2237.
- [2] S. Ayu K. Gitadewi and Luh Gede Meydianawathi, "Determinants of Coffee Shop Business Income in Kintamani District," *International Journal of Economics, Management and Accounting*, vol. 2, no. 2, pp. 368–377, Apr. 2025, doi: 10.61132/ijema.v2i2.656.
- [3] E. Yulianti Faridah and R. S. Dewi, "Effect of Proximity on The Layout and Selection of Chair Types in The Coffee Shop Industry," *Journal of Industrial Engineering and Halal Industries*, vol. 5, no. 1, pp. 1–10, Jul. 2024, doi: 10.14421/jieh.4648.
- [4] O. Tanvir, R. Adeel, and K. Masood, "Role of Data Analytics, Business Intelligence, and Performance Management in Enhancing Strategic Marketing Decision-Making," *Qlantic Journal of Social Sciences*, vol. 5, no. 4, pp. 445–453, Dec. 2024, doi: 10.55737/qjss.vi-i.25294.
- [5] M. R. Prasetyawan and M. Wasesa, "Business Escalation Strategy Using Time Series Forecasting for Hotel X in Yogyakarta," *International Journal of Current Science Research and Review*, vol. 06, no. 08, Aug. 2023, doi: 10.47191/ijcsrr/V6-i8-27.
- [6] Erik JahnKarthik KarnikLeonard J. Schulman, "Causal Identification in Time Series Models," 2025.
- [7] Z. Dong and H. Hao, "Vegetable Sales Forecasting and Pricing Strategy Planning Based on ARIMA Algorithm and Linear Programming Model," in *2024 4th International Symposium on Computer Technology and Information Science (ISCTIS)*, IEEE, Jul. 2024, pp. 376–381. doi: 10.1109/ISCTIS63324.2024.10698868.
- [8] A. Gupta and A. Kumar, "Mid Term Daily Load Forecasting using ARIMA, Wavelet-ARIMA and Machine Learning," in *2020 IEEE International Conference on Environment and Electrical Engineering and 2020 IEEE Industrial and Commercial Power Systems Europe (EEEIC / I&CPS Europe)*, IEEE, Jun. 2020, pp. 1–5. doi: 10.1109/EEEIC/ICPSEurope49358.2020.9160563.
- [9] S. Yang, Z. Xu, and K. Ren, "An Improved Parameter Extraction Optimization Algorithm for RF Devices," *Micromachines (Basel)*, vol. 16, no. 4, p. 432, Apr. 2025, doi: 10.3390/mi16040432.
- [10] S. Mukhanov, S. Amirgaliyeva, D. Amrin, S. Zhakypbekov, and B. Amirgaliyev, "COMPARISON OF SINGLE AND HYBRID NEURAL NETWORK MODELS FOR STOCK PRICE FORECASTING," *Вестник КазАТК*, vol. 136, no. 1, pp. 402–418, Feb. 2025, doi: 10.52167/1609-1817-2025-136-1-402-418.
- [11] M. Yeasin, K. N. Singh, R. V. R. K. Paul, and A. Lama, "Application of SARIMA model for precipitation modelling driven by exogenous variables," *MAUSAM*, vol. 76, no. 2, pp. 365–372, Mar. 2025, doi: 10.54302/mausam.v76i2.5487.
- [12] N. W. Marbun, F. A. Iz, M. Ramadhan, L. J. H. Kogoya, L. F. Nugraha, and U. Nugraha, "Payment and Transaction Risk Management at Coffeeshop X Using NIST 800-30 Framework," *JUSTINFO | Jurnal Sistem Informasi dan Teknologi Informasi*, vol. 1, no. 2, pp. 135–146, Jun. 2024, doi: 10.33197/justinfo.vol1.iss2.2023.1745.
- [13] T. Stoilov, K. Stoilova, and L. Iliev, "Inventory Optimization with Several Resources," *WSEAS TRANSACTIONS ON BUSINESS AND ECONOMICS*, vol. 22, pp. 151–156, Jan. 2025, doi: 10.37394/23207.2025.22.14.